

SWAPNIL MALLICK

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EDUCATION

The University of British Columbia, Vancouver

Doctor of Philosophy, Electrical and Computer Engineering

Sep 2025 - Present

Overall Grade: 88/100 (A)

University of Southern California, Los Angeles

Master of Science, Computer Science

Aug 2021 - Dec 2023

Overall GPA: 3.56/4

Maulana Abul Kalam Azad University Of Technology (WBUT), India

Bachelor of Technology, Computer Science and Engineering

Jun 2016 - Jul 2020

Overall GPA: 9.03/10

WES Converted GPA: 3.96/4

RESEARCH EXPERIENCES

The University of North Carolina at Chapel Hill

Spatial & Physical Intelligence (SPIN) Lab

Remote

March 2024 - January 2025

- Collaborators: Prof. Roni Sengupta (Google Scholar: <https://bit.ly/4ejIRdC>) and Jun Myeong Choi (Google Scholar: <http://bit.ly/44Ls6Hk>)
- Occasionally collaborated on detecting student video lecture engagement with passive monitor illumination
- Implemented a personalized ViT backbone architecture which shows an improvement in performance (74.95%) when compared with the baseline architecture (~ 71%) in dark lighting conditions
- Experimented with diffusion models and SDS loss (per-image optimization) to improve the monitor image prediction quality
- Explored the application of self and cross attention based architectures in detecting video engagement with the ultimate goal of building a generalized model for the task

University of Southern California

USC CPS-VIDA (Cyber-Physical Systems - Verification, Intelligence, Design, and Analysis)

Los Angeles, California

March 2024 - January 2025

- Supervisor: Prof. Jyotirmoy Vinay Deshmukh (Google Scholar: <https://bit.ly/3UIaIxd>)
- Co-authored a research paper on formal and sampling-based techniques for automatic generation of edge-case scenarios in safety-critical autonomous systems, with applications in autonomous driving and human-robot collaboration in industrial settings (**RV '24**)
- Familiarized myself with Scenic and variants of Rapidly-exploring Random Trees (RRT's)
- Broadly explored the application of generative models like, VQGAN, diffusion models for generating safety-critical scenarios for comprehensive self-driving car testing

University of Southern California

USC CPS-VIDA (Cyber-Physical Systems - Verification, Intelligence, Design, and Analysis)

Los Angeles, California

textitSupervisor: Dr. Jyotirmoy Vinay Deshmukh (Associate Professor)

May 2023 - August 2023

- Collaborators: Prof. Jyotirmoy Vinay Deshmukh (Google Scholar: <https://bit.ly/3UIaIxd>), Shuvam Ghosal (Google Scholar: <https://bit.ly/4dBdcWE>) and Anand Balakrishnan (Google Scholar: <https://bit.ly/4364Gv5>)
- Developed a safety monitoring framework for detecting the robustness of existing pedestrian detection models in low-light and adverse weather conditions using Timed Quality Temporal Logic (TQTL) (**RV '23**)
- Compared the robustness of vanilla YOLO and IA-YOLO in both foggy as well as night conditions
- Found IA-YOLO to perform better as compared to vanilla YOLO in both foggy as well as night conditions
- Experiments indicated that IA-YOLO was able to detect pedestrians successfully in both foggy and night conditions whereas vanilla YOLO was able to detect pedestrians reasonably in night conditions but failed to satisfy the TQTL specifications in foggy weather conditions
- Proposed TQTL as a quality metric that will help in debugging or improving the existing pedestrian detection models

Indian Statistical Institute

Computer Vision And Pattern Recognition (CVPR) Unit

Kolkata, India

Certificate Link: <https://bit.ly/3S0wxYo>

July 2019 - May 2020

- Supervisor: Prof. Ujjwal Bhattacharya (Google Scholar: <https://bit.ly/3CsdIaH>)
- Designed three separate architectures for detecting handwritten Bengali numerals - one mimicking the LeNet-5 network and the other two using Residual and Inception blocks
- Employed DCGAN to create additional samples to deal with the initial scarcity of training data
- Found that the simpler LeNet-5 architecture achieved a higher accuracy of ~ 98.2% on test data as compared to the architectures that used Inception block (~ 97.4%) and Residual block (~ 97.1%)

PUBLICATIONS

- C2. Vidisha Kudalkar, Navid Hashemi, Shilpa Mukhopadhyay, **Swapnil Mallick**, Christof Budnik, Parinitha Nagaraja, and Jyotirmoy V. Deshmukh. [Sampling-Based and Gradient-Based Efficient Scenario Generation](#). 24th International Conference on Runtime Verification, pp. 70-88. Springer Nature Switzerland, 2024.
DOI: https://doi.org/10.1007/978-3-031-74234-7_5 || PDF Link: <https://bit.ly/48Plesq>
- C1. **Swapnil Mallick***, Shuvam Ghosal*, Anand Balakrishnan, Jyotirmoy Deshmukh. [Safety monitoring for pedestrian detection in adverse conditions](#). 23rd International Conference on Runtime Verification, pp. 389-399. Springer Nature Switzerland, 2023.
DOI: https://doi.org/10.1007/978-3-031-44267-4_22 || PDF Link: <https://par.nsf.gov/servlets/purl/10495152>

ACADEMIC PROJECTS

BrickRL: Reinforcement Learning for Automated Brick Placement

Project Code Repository: <https://bit.ly/3RetM7x>

April 2026

Project Report: <https://bit.ly/49Q1WqW>

- This study, conducted as a course project for EECE 571.F at UBC, was an attempt to see whether standard Reinforcement Learning algorithms can be used to automate the process of brick placement along defined target lines
- Developed a Gymnasium-style 2-D simulation environment used as the testbed for realizing the potential of Reinforcement Learning algorithms to learn correct brick placement policies
- The simple 2-D grid world had a default dimension of 100 x 100, an inventory of bricks on the top-left corner with each brick having a dimension of 5 x 2
- The target geometric shape along which the agent had to place the bricks was represented by one or more straight lines
- Ran experiments on three types of geometric target shapes - i) a horizontal line; ii) a vertical line; iii) an L-shaped target consisting of a horizontal and a vertical line
- Trained standard RL algorithms - PPO, TRPO and A2C - in simulation to study whether the agent is able to interact with the environment and learn to place all the bricks from the inventory to the target locations along the target line(s)
- Studies showed that none of the algorithms were able to learn correct and complete brick placement policies along any of the target shapes
- Overall, A2C was found to attain the highest completion rates on each of the target wall types - 66.7% for horizontal, 75% for vertical and 60% for the L-shaped target
- PPO performed reasonably on simpler targets achieving completion rates of 66.7% and 62.5% on the horizontal and vertical targets but failed in the L-shaped target achieving only a completion rate of 10%
- TRPO performed poorly on all three targets achieving completion rates of 50%, 37.5% and 10% on horizontal, vertical and L-shaped targets respectively

Analysing Twitter Data on Rhino Conservation using Natural Language Processing

Project Report: <https://bit.ly/4eJ5mcX>

November 2023

- This study, conducted as group project for CSCI 461 at USC, was designed to demonstrate the utility of social media data in monitoring public opinion regarding illegal rhino poaching and trade
- Various toolkits, including Pattern, TextBlob, VADER, and Twitter-roBERTa-base, were utilized to gauge the sentiment expressed in the tweets; used SHAP values to interpret the behavior of the models/toolkits
- In parallel, topic modeling techniques such as LDA and BTM were applied to gain deeper insights into the prevalent discussion topics and identify the user categories participating in these conversations
- The exploratory data analysis conducted on rhino-related tweets revealed that the volume of such tweets exhibited a comparative increase from 2014 to 2019, reaching a peak in 2018
- For assessing the sentiment of tweets in conservation-related discussions, the results obtained did not provide a clear indication of the effectiveness of relying on pretrained NLP tools
- The topic modeling analysis revealed a dual nature of discussions on rhinos, encompassing both general themes such as 'Poaching' and 'Horn/Ivory Trade', as well as event-specific topics like the 'Last White Male Northern Rhino Extinction' and 'Horn Stealing/Veterinarian'

CHIP: Contrastive Hierarchical Image Pretraining

Project Report: <https://bit.ly/4en84EA>

May 2023

- This study, conducted in group of four students as the course project of CSCI 566 at USC, proposed a few-shot three-level hierarchical contrastive loss-based classification model that can classify an object of any unseen animal class into a relatively general category in a hierarchically based classification
- The proposed model was trained on a subset of the ImageNet (ILSVRC-12) dataset containing only the animal classes (366 classes in total) with 1300 images in each class
- The trained model was evaluated on a dataset created by us that contained 20 unseen animal classes not present in the ImageNet dataset
- The trained model, evaluated on 20 unseen animal classes, could classify 14 of them to their correct parent and super parent class

Adaptive Cruise Control System of Self-driving Car

Project Code Repository: <https://bit.ly/48UcXTI>

December 2022

- Designed an adaptive cruise control system of a self-driving car using YOLOv5 model for object detection and Monodepth2 for distance estimation

- The Monodepth2 model was used to output a disparity map of the bounding box region of the detected lead car from which the value at the centroid position of the bounding box was taken to get the estimated distance
- This was then fed to the PID controller of the ego car, which has to maintain a safe distance to the lead car
- The system was tested in CARLA simulator Town 6 and the ego car was found to always maintain a safe distance to the lead car

Stock Price Prediction of a Company using LSTM

Project Code Repository: <https://bit.ly/3MOREpN>

June 2022

- Leveraged an LSTM architecture to predict the stock price of a company
- Measured the performance of the model using RSME score
- Received highest RMSE score of 7.25 on unknown test data after running five times

A Multimodal Approach to Speech Emotion Recognition

Project Report: <https://bit.ly/4gKNMqD> | Project Code Repository: <https://bit.ly/3BuQu3u>

May 2022

- This study, conducted as the course project of CSCI 544 at USC, uses audio and text as modalities from the IEMOCAP dataset for human emotion recognition
- A VGG frame architecture was trained from scratch to detect emotions from audio and a pretrained BERT was finetuned to predict the emotions from text transcriptions
- The text-only and audio-only architectures were combined to design a multimodal architecture for emotion recognition
- The multimodal architecture achieved a considerably higher accuracy of 76.9% as compared to the models that used either text (72.4%) or audio (47.8%) as the modality

Cipher Text Prediction using BiLSTM

Project Code Repository: <https://bit.ly/3QiJ5Jw>

March 2022

- Designed a binary classifier that can classify whether a piece of text is encoded or not
- Converted text into word embeddings and employed BiLSTM for classification task
- The classifier achieved ~ 87.5% accuracy on training data and ~ 85% accuracy on test data

COURSEWORK

Graduate Courses (PhD):

- **EECE 571.S:** Machine Teaching for Robotics (Link to Syllabus: <https://bit.ly/4uKZaca>)
- **EECE 571.F:** Advanced Topics in Deep Learning (Link to Syllabus: <https://bit.ly/4dKxvCu>)

Graduate Courses (Master's):

- **CSCI 513:** Autonomous Cyber-Physical Systems (**Scored 476.05 out of 440**)
 - **Link to Blackboard grade sheet:** <https://bit.ly/4ePrHZB>
 - **Link to Course Syllabus:** <https://bit.ly/3YAvKyZ>
- **CSCI 585:** Database Systems (**Scored 100 out of 100**)
 - **Link to D2L grade sheet:** <https://bit.ly/4urQrvC>
- **CSCI 461:** AI for Sustainable Development (**Scored 147.50 out of 127**)
 - **Link to Blackboard grade sheet:** <https://bit.ly/494nQnE>
- **CSCI 567:** Machine Learning
- **CSCI 566:** Deep Learning and its Applications
- **CSCI 544:** Applied NLP
- **CSCI 572:** Information Retrieval and Web Search Engines
- **CSCI 561:** Foundations of AI
- **CSCI 570:** Analysis of Algorithms

Undergraduate Courses:

- **CS 402:** Formal Language and Automata Theory
- **CS 403:** Computer Architecture
- **CS 502:** Microprocessors and Microcontrollers
- **CS 503:** Discrete Mathematics
- **CS 504D:** Object Oriented Programming
- **CS 591:** Design and Analysis of Algorithm
- **CS 601:** Database Management System
- **CS 602:** Computer Networks
- **CS 603:** Operating System
- **CS 604A:** Information and Coding Theory
- **CS 605A:** Operations Research
- **CS 701:** Software Engineering
- **CS 702:** Compiler Design
- **CS 703C:** Artificial Intelligence
- **CS 801D:** Cryptography and Network Security

SKILLS

- **Programming:** Python, C, C++
- **ML Frameworks:** Keras, TensorFlow, PyTorch, scikit-learn
- **Data Visualization:** Matplotlib, seaborn
- **Database:** MySQL, PostgreSQL, MongoDB
- **Software Tools:** Anaconda, Jupyter Notebook, L^AT_EX, CARLA Simulator, KNIME, WEKA, RapidMiner, Tableau, Gurobi Optimization

CERTIFICATIONS

- Deep Learning Specialization (DeepLearning.AI) (Link: <https://bit.ly/48EgkOS>)
 - ☐ Neural Networks and Deep Learning (Link: <https://bit.ly/3O1M5YN>)
 - ☐ Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization (Link: <https://bit.ly/48E1w2C>)
 - ☐ Structuring Machine Learning Projects (Link: <https://bit.ly/48Kg3tJ>)
 - ☐ Convolutional Neural Networks (Link: <https://bit.ly/4fgHYUJ>)
 - ☐ Sequence Models (Link: <https://bit.ly/40Bl0D4>)
- Natural Language Processing Specialization (DeepLearning.AI) (Link: <https://bit.ly/3AzIRsm>)
 - ☐ Natural Language Processing with Classification and Vector Spaces (Link: <https://bit.ly/3YDBAzB>)
 - ☐ Natural Language Processing with Probabilistic Models (Link: <https://bit.ly/3AxSkAj>)
 - ☐ Natural Language Processing with Sequence Models (Link: <https://bit.ly/3ULl6V1>)
 - ☐ Natural Language Processing with Attention Models (Link: <https://bit.ly/40CoFkj>)
- MATLAB (Vanderbilt University) (Link: <https://bit.ly/3YZZKGc>)
- Bayesian Statistics (UC Santa Cruz) (Link: <https://bit.ly/3Axq7cU>)
- Introduction to Big Data (UC San Diego) (Link: <https://bit.ly/48E5590>)
- Big Data Integration and Processing (UC San Diego) (Link: <https://bit.ly/3AyiDq8>)
- Big Data Modeling and Management Systems (UC San Diego) (Link: <https://bit.ly/3YHNGYw>)
- Machine Learning with Big Data (UC San Diego) (Link: <https://bit.ly/40DPFzN>)
- Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning (DeepLearning.AI) (Link: <https://bit.ly/40CTV2p>)
- Convolutional Neural Networks in TensorFlow (DeepLearning.AI) (Link: <https://bit.ly/4fEZVMf>)

BLOG POSTS

- A Survey On Data Driven Approaches To Scenario Generation For Autonomous Vehicle Testing
Link: <https://swapnilmallick.github.io/blogs/scenario-generation.html>
- A Guide To Generative Adversarial Network
Link: <https://swapnilmallick.github.io/blogs/gan.html>